

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0521

Longdon 138 kV
November 27, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0521. The Interconnection Customer proposes a 218.7 MW wind facility interconnecting at the Longdon 138 kV substation in the Kentucky area. The request was received on September 25, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 179 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$39,863,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Terry Co-Cedar Tap 138 KV L, the Casino-Rapids Energy Ctr 138 KV L loads to 124.2% of its 239 MVA rating. Project DFAX: 56.0%.

Barnabie Cable Sta-Casino 138 KV: 124.1% loading (248 MVA) following the outage of the Cedar Tap-Terry Co Substation 138 KV L; DFAX = 56.0%.

Campbell Jct-Rapids Energy Ctr 138 KV: 119.5% loading (230 MVA) following the outage of the Cedar Tap-Terry Co 138 KV; DFAX = 56.0%.

Skeena-Terry Co 138 KV line: 115.2% loading (250 MVA) following the outage of the Barnabie Cable Sta-Casino 138 KV line; DFAX = 49.5%.

For the loss of Wixom-Albion Rd 138 KV, the Anso N-Pin Hook 138 KV loads to 115.1% of its 191 MVA rating. Project DFAX: 21.1%.

Cedar Tap Substation-Terry Co 138 KV L: 114.8% loading (254 MVA) following the outage of the Casino-Barnabie Cable Sta 138 KV Line; DFAX = 49.5%.

Albion Road Substation-Platte City 138 KV Line: 106.2% loading (238 MVA) following the outage of the Albion Rd-Central Campus Unm 138 KV line; DFAX = 20.1%.

For the loss of Moss Bluff-Barnabie Cable Sta 138 KV, the Barnabie Cable Sta-Longdon 138 KV loads to 106.0% of its 217 MVA rating. Project DFAX: 68.9%.

For the loss of Wixom-Albion Road 138 KV line, the Pin Hook-Skeena 138 KV loads to 105.6% of its 218 MVA rating. Project DFAX: 21.1%.

The Platte City-C-33 Doe 138 KV is overloaded to 103.2% of its 234 MVA emergency rating for the loss of the Central Campus Unm-Albion Rd Substation 138 KV. The project contributes 20.1% DFAX (44.1 MW) to this overload.

For the loss of Albion Rd-Central Campus Unm 138 KV L, the Sand Springs-C-33 Doe 138 KV line loads to 101.4% of its 229 MVA rating. Project DFAX: 20.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Casino-Rapids Energy Ctr 138 KV L	Rebuild line with higher-capacity conductor	\$3,754,000	66% (\$2,470,000)
Barnabie Cable Sta-Casino 138 KV	Rebuild line with higher-capacity conductor	\$13,727,000	66% (\$9,033,000)
Campbell Jct-Rapids Energy Ctr 138 KV	Rebuild line with higher-capacity conductor	\$3,281,000	66% (\$2,159,000)
Skeena-Terry Co 138 KV L	Rebuild line with higher-capacity conductor	\$13,803,000	65% (\$8,946,000)
Pin Hook-Anson North 138 KV	Rebuild line with higher-capacity conductor	\$18,169,000	24% (\$4,372,000)
Cedar Tap-Terry Co 138 KV	Rebuild line with higher-capacity conductor	\$10,892,000	65% (\$7,059,000)
Platte City-Albion Rd 138 KV Line	Reconductor line	\$4,951,000	22% (\$1,092,000)
Longon-Barnabie Cable Sta 138 KV	Reconductor line	\$2,981,000	100% (\$2,981,000)
Pin Hook-Skeena 138 KV	Reconductor line	\$3,242,000	24% (\$780,000)
Platte City-C-33 Doe 138 KV	Reconductor line	\$1,605,000	22% (\$354,000)
C-33 Doe-Sand Springs 138 KV L	Reconductor line	\$2,799,000	22% (\$617,000)
	Total allocated to project		\$39,863,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.